1. Database Changes

SQL

```
CREATE TABLE IF NOT EXISTS
customers (     id INT UNSIGNED
AUTO_INCREMENT PRIMARY KEY,
full_name VARCHAR(150) NOT NULL,
email VARCHAR(150) NOT NULL,
created_at DATETIME NULL,
updated_at DATETIME NULL );

CREATE TABLE IF NOT EXISTS users (
id INT UNSIGNED AUTO_INCREMENT
PRIMARY KEY,     username
VARCHAR(80) NOT NULL UNIQUE,
full_name VARCHAR(150) NOT NULL,
avatar VARCHAR(255) NULL,
created_at DATETIME NULL,
updated_at DATETIME NULL );

-- If your users table already exists, run only this line when needed:
ALTER TABLE users ADD COLUMN avatar VARCHAR(255) NULL AFTER full_name;

-- Example account using the submitted profile photo
INSERT INTO users (username, full_name, avatar, created_at, updated_at)
VALUES ('aldrichsy', 'Aldrich Sy', 'aldrich_sy.png', NOW(), NOW());

INSERT INTO users (username, full_name, avatar, created_at, updated_at)
VALUES ('angelamae', 'Angela Mae', 'angela_mae.png', NOW(), NOW());
```

Public Upload Directory

```
public/
uploads/
avatars/
placeholder.png
aldrich_sy.png
angela_mae.png
```

Create the folder public/uploads/avatars. Put a placeholder image named placeholder.png inside it. Also copy the provided aldrich_sy.png and angela_mae.png files into the same folder.

2. Routes app/Config/Routes.php

```
$routes->get('/customers', 'Customers::index');
$routes->get('/customers/new', 'Customers::new');
$routes->post('/customers/create', 'Customers::create');
$routes->get('/customers/edit/(:num)', 'Customers::edit/$1');
$routes->post('/customers/update/(:num)', 'Customers::update/$1');

$routes->get('/users', 'Users::index');
$routes->get('/users/new', 'Users::new');
$routes->post('/users/create', 'Users::create');
$routes->get('/users/edit/(:num)', 'Users::edit/$1');
$routes->post('/users/update/(:num)', 'Users::update/$1');
```

3. Models app/Models/CustomerModel.php

```
<?php namespace

App\Models; use

CodeIgniter\Mode

l;

class CustomerModel extends Model
{
    protected $table = 'customers';
protected $primaryKey = 'id';
protected $allowedFields =
```

```
['full_name', 'email'];     protected
$useTimestamps = true; }
```

app/Models/UserModel.php

```
<?php namespace

App\Models; use

CodeIgniter\Mode

l;

class UserModel extends Model
{
    protected $table =
'users';     protected
$primaryKey = 'id';
    protected $allowedFields = ['username', 'full_name',
'avatar'];     protected $useTimestamps = true;
}
```

4. Customer Controller

app/Controllers/Customers.php

```
<?php
namespace

App\Controllers; use

App\Models\CustomerMo

del;

class Customers extends BaseController
{
    protected $customers;

    public function __construct()
    {
        helper(['form', 'url']);
        $this->customers = new CustomerModel();
    }

    public function index()
    {
        return view('customers/index', [
            'customers' => $this->customers->orderBy('id', 'DESC')->findAll()
        ]);
    }

    public function new()
    {
        return
view('customers/new');     }

    public function create()
    {
        $rules = [
            'full_name' => 'required|min_length[2]|max_length[150]',
            'email'     => 'required|valid_email|max_length[150]'
        ];

        if (!$this-
>validate($rules)) {
return redirect()->back()-
>withInput()
                ->with('errors', $this->validator->getErrors());
        }

        $this->customers->insert([
            'full_name' => trim($this->request->getPost('full_name')),
            'email'     => trim($this->request->getPost('email'))
        ]);

        return redirect()->to('/customers')->with('success', 'Customer
created successfully.');     }

    public function edit($id)
    {
        $customer = $this-
>customers->find($id);         if
(!$customer) {
            throw \CodeIgniter\Exceptions\PageNotFoundException::forPageNotFound();
        }
        return view('customers/edit',
['customer' => $customer]);     }

    public function update($id)
    {
```

```
        $customer = $this->customers->find($id);         if (!$customer) {
            throw \CodeIgniter\Exceptions\PageNotFoundException::forPageNotFound();         }

        $rules = [
            'full_name' => 'required|min_length[2]|max_length[150]',
'email'     => 'required|valid_email|max_length[150]'
        ];

        if (!$this->validate($rules)) {
return redirect()->back()->withInput()
                ->with('errors', $this->validator->getErrors());
        }

        $this->customers->update($id, [
            'full_name' => trim($this->request->getPost('full_name')),
            'email'     => trim($this->request->getPost('email'))
        ]);

        return redirect()->to('/customers')->with('success', 'Customer updated successfully.');
    }
}
```

5. User Controller with Avatar Validation and Preparation app/Controllers/Users.php

```
<?php
namespace App\Controllers;

use App\Models\UserModel; use Config\Services;

class Users extends BaseController
{
    protected $users;

    public function __construct()
    {
        helper(['form', 'url']);
        $this->users = new UserModel();
    }

    public function index()
    {
        return view('users/index', [
            'users' => $this->users->orderBy('id', 'DESC')->findAll()
        ]);
    }

    public function new()
    {
        return view('users/new');     }

    public function create()
    {
        $rules = [
            'username'  => 'required|min_length[3]|max_length[80]|is_unique[users.username]',
            'full_name' => 'required|min_length[2]|max_length[150]'
        ];

        if (!$this->validate($rules)) {
            return redirect()->back()->withInput()
```

```
                ->with('errors', $this->validator->getErrors());
        }

        $this->users->insert([
            'username'  => trim($this->request-
>getPost('username')),              'full_name' =>
trim($this->request->getPost('full_name'))
        ]);

        return redirect()->to('/users')->with('success', 'User
created successfully.');     }

    public function edit($id)
    {
        $user = $this->users-
>find($id);         if
(!$user) {
            throw \CodeIgniter\Exceptions\PageNotFoundException::forPageNotFound();
        }
        return view('users/edit',
['user' => $user]);     }

    public function update($id)
    {
        $user = $this->users-
>find($id);         if
(!$user) {
            throw
\CodeIgniter\Exceptions\PageNotFoundException::forPageNotFoun
d();         }

        $rules = [
            'username'  =>
"required|min_length[3]|max_length[80]|is_unique[users.username,id,{$id}]",
            'full_name' => 'required|min_length[2]|max_length[150]',
            'avatar'    =>
'if_exist|max_size[avatar,2048]|is_image[avatar]|mime_in[avatar,image/jpg,image/jpeg,image/png]|ext_
in[avatar,
        ];

        if (!$this->validate($rules)) {
            return redirect()->back()->withInput()
                ->with('errors', $this->validator->getErrors());
        }

        $data = [
            'username'  => trim($this->request-
>getPost('username')),              'full_name' =>
trim($this->request->getPost('full_name'))
        ];

        $file = $this->request->getFile('avatar');
        if ($file && $file->isValid() && !$file->hasMoved()) {
            $newName = $file->getRandomName();
            $uploadPath = FCPATH . 'uploads/avatars/';
            $file->move($uploadPath, $newName);

            $image = Services::image()
                ->withFile($uploadPath . $newName)
                ->fit(300, 300, 'center')
                ->save($uploadPath . $newName, 85);

            if (!empty($user['avatar']) && $user['avatar'] !== 'placeholder.png') {
                $old = $uploadPath .
$user['avatar'];                 if
(is_file($old)) @unlink($old);
}

            $data['avatar'] = $newName;
```

```
        }

        $this->users->update($id, $data);
        return redirect()->to('/users')->with('success', 'User updated successfully.');
    }
}
```

6. Customer Views app/Views/customers/new.php

```
<!doctype html>
<html>
<head><title>New Customer</title></head>
<body>
<h2>New Customer</h2>

<?php if (session('errors')): ?>
    <ul style="color:red;">
        <?php foreach (session('errors') as $error): ?>
            <li><?= esc($error) ?></li>
        <?php endforeach; ?>
    </ul>
<?php endif; ?>

<form method="post" action="<?= site_url('/customers/create') ?>">
    <?= csrf_field() ?>
    <label>Full Name</label><br>
    <input type="text" name="full_name" value="<?= old('full_name') ?>" required><br><br>

    <label>Email</label><br>
    <input type="email" name="email" value="<?= old('email') ?>" required><br><br>

    <button type="submit">Save Customer</button>
</form>
</body>
</html>
```

app/Views/customers/edit.php

```
<!doctype html>
<html>
<head><title>Edit Customer</title></head>
<body>
<h2>Edit Customer</h2>

<?php if (session('errors')): ?>
    <ul style="color:red;">
        <?php foreach (session('errors') as $error): ?>
            <li><?= esc($error) ?></li>
        <?php endforeach; ?>
    </ul>
<?php endif; ?>

<form method="post" action="<?= site_url('/customers/update/' . $customer['id']) ?>">
    <?= csrf_field() ?>
    <label>Full Name</label><br>
    <input type="text" name="full_name"
           value="<?= old('full_name', $customer['full_name']) ?>" required><br><br>

    <label>Email</label><br>
    <input type="email" name="email"
           value="<?= old('email', $customer['email']) ?>" required><br><br>

    <button type="submit">Update Customer</button>
</form>
</body>
</html>
```

app/Views/customers/index.php

```
<!doctype html>
```

```
<html>
<head><title>Customers</title><
/head>
<body>
<h2>Customers</h2> <a href="<?=
site_url('/customers/new') ?>">+ New
Customer</a>

<?php if (session('success')): ?>
<p style="color:green;"><?= esc(session('success')) ?></p>
<?php endif; ?>

<table border="1" cellpadding="8" cellspacing="0">
<tr><th>ID</th><th>Full
Name</th><th>Email</th><th>Action</th></tr> <?php
foreach ($customers as $customer): ?>
<tr>
    <td><?= esc($customer['id']) ?></td>
    <td><?= esc($customer['full_name']) ?></td>
    <td><?= esc($customer['email']) ?></td>
    <td><a href="<?= site_url('/customers/edit/' . $customer['id']) ?>">Edit</a></td>
</tr>
<?php endforeach; ?>
</table>
</body>
</html>
```

7. User Views app/Views/users/new.php

```
<!doctype html>
<html>
<head><title>New User</title></head>
<body>
<h2>New User</h2>

<?php if (session('errors')): ?>
    <ul style="color:red;">
        <?php foreach (session('errors') as $error): ?>
            <li><?= esc($error) ?></li>
        <?php endforeach; ?>
    </ul>
<?php endif; ?>

<form method="post" action="<?= site_url('/users/create') ?>">
    <?= csrf_field() ?>
    <label>Username</label><br>
    <input type="text" name="username" value="<?= old('username') ?>" required><br><br>

    <label>Full Name</label><br>
    <input type="text" name="full_name" value="<?= old('full_name') ?>" required><br><br>

    <button type="submit">Save User</button>
</form>
</body>
</html>
```

app/Views/users/edit.php

```
<!doctype html>
<html>
<head><title>Edit User</title></head>
<body>
<h2>Edit User</h2>

<?php if (session('errors')): ?>
    <ul style="color:red;">
        <?php foreach (session('errors') as $error): ?>
            <li><?= esc($error)
?></li>         <?php
endforeach; ?>
    </ul>
<?php endif; ?>

<?php $avatar = !empty($user['avatar']) ? $user['avatar'] :
'placeholder.png'; ?> <img src="<?=
base_url('uploads/avatars/' . $avatar) ?>" width="100"
height="100"      style="object-fit:cover;border-radius:50%;"
alt="Avatar"><br><br>

<form method="post" enctype="multipart/form-data"
      action="<?= site_url('/users/update/' .
$user['id']) ?>">     <?= csrf_field() ?>

    <label>Username</label><br>
    <input type="text" name="username"
           value="<?= old('username', $user['username']) ?>" required><br><br>

    <label>Full Name</label><br>
    <input type="text" name="full_name"
           value="<?= old('full_name', $user['full_name']) ?>" required><br><br>

    <label>Profile Picture (JPG/PNG, max 2 MB)</label><br>
    <input type="file" name="avatar" accept=".jpg,.jpeg,.png,image/jpeg,image/png"><br><br>

    <button type="submit">Update User</button>
</form>
</body>
```

```
</html>
```

app/Views/users/index.php

```
<!doctype html>
<html> <head><title>User
Accounts</title></head>
<body>
<h2>User Accounts</h2> <a href="<?=
site_url('/users/new') ?>">+ New
User</a>

<?php if (session('success')): ?>
<p style="color:green;"><?= esc(session('success')) ?></p>
<?php endif; ?>

<table border="1" cellpadding="8" cellspacing="0">
<tr><th>Avatar</th><th>Username</th><th>Full Name</th><th>Action</th></tr>
<?php foreach ($users as $user): ?>
<?php $avatar = !empty($user['avatar']) ? $user['avatar'] : 'placeholder.png'; ?>
<tr>
    <td>
        <img src="<?= base_url('uploads/avatars/' . $avatar) ?>"
             width="60" height="60" style="object-fit:cover;border-radius:50%;" alt="Avatar">
    </td>
    <td><?= esc($user['username']) ?></td>
    <td><?= esc($user['full_name']) ?></td>
    <td><a href="<?= site_url('/users/edit/' . $user['id']) ?>">Edit</a></td>
</tr>
<?php endforeach; ?>
</table>
</body>
</html>
```

8. Expected Browser Outputs

New Customer
Full Name Juan Dela Cruz
Email juan@example.com
Save Customer

New User
Username angelamae
Full Name Angela Mae
Save User

Edit User
Current Avatar Angela Mae profile photo
Username angelamae
Full Name Angela Mae
Profile Picture angela_mae.png - JPG/PNG, maximum 2 MB
Update User

User Accounts Listing Output

Avatar	Username	Full Name	Action
	aldrichsy	Aldrich Sy	Edit
	angelamae	Angela Mae	Edit